CASO PRÁCTICO 2

Se pide:
Acceder a la siguiente página web.
Y descargar el fichero de deuda de ayuntamientos del año 2016.

Una vez descargado y abierto para analizar su información, resolver los siguientes ejercicios utilizando Hive:

1 - Indicar el comando de creación de una tabla en Hive que contenga la información de la pestaña *Datos* con las siguientes columnas:

Anio
Cod_Comunidad
Comunidad
Cod_Provincia
Provincia
Cod_corp
Corporacion
Deuda

2 - Cargar el fichero anterior descargado en esa tabla e indicar el comando empleado.

3 - Obtener la corporación que tiene la tercera deuda más grande.

4 - Obtener la corporación que tiene la deuda más grande de Andalucía.

5 - Obtener la suma de deudas de cada comunidad autónoma, ordenadas por comunidad.

6 - Obtener la suma de las deudas de las corporaciones que empiezan por vocal.

7 - Exportar las corporaciones cuya deuda sea mayor a 10 000 y pertenezcan a Cataluña, a un fichero llamado “Deudas”, y adjuntarlo. Se puede emplear un comando similar al siguiente:

```
INSERT OVERWRITE LOCAL DIRECTORY
'/home/bigdata/ejemplosHive/RESULTADO/' row format delimited
fields terminated by ',' select * from .........
```

Solución:

Antes de nada, lo primero que hacemos es comprobar las versiones de **Java** y **Hadoop**.

```
bigdata@bigdata-IMF:~$ java -version
openjdk version "1.8.0_312"
OpenJDK Runtime Environment (build 1.8.0_312-8u312-b07-0ubuntu1~20.04-b07)
OpenJDK 64-Bit Server VM (build 25.312-b07, mixed mode)
bigdata@bigdata-IMF:~$ hadoop version
Hadoop 3.3.1
Source code repository https://github.com/apache/hadoop.git -r a3b9c37a397ad4188041dd80621bdeefc46885f2
Compiled by ubuntu on 2021-06-15T05:13Z
Compiled with protoc 3.7.1
From source with checksum 88a4ddb2299aca054416d6b7f81ca55
This command was run using /home/bigdata/hadoop/share/hadoop/common/hadoop-common-3.3.1.jar
bigdata@bigdata-IMF:~$
```

Actualizamos el índice local de paquetes para que el sistema sepa cual es la última versión disponible de **curl** (la herramienta que utilizaremos para descargar los ficheros del enlace) y **xlsx2csv** (la herramienta que utilizaremos para transformar el .xlsx en .csv).

```
bigdata@bigdata-IMF:~$ sudo apt update
[sudo] contraseña para bigdata:
Des:1 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Obj:2 http://es.archive.ubuntu.com/ubuntu focal InRelease
Des:3 http://es.archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Des:4 http://es.archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]
Des:5 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [3.564 kB]
Des:6 http://es.archive.ubuntu.com/ubuntu focal-updates/main i386 Packages [1.114 kB]
Des:7 http://es.archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [3.954 kB]
Des:8 http://security.ubuntu.com/ubuntu focal-security/main i386 Packages [881 kB]
Des:9 http://security.ubuntu.com/ubuntu focal-security/main Translation-en [518 kB]
Des:10 http://security.ubuntu.com/ubuntu focal-security/main amd64 DEP-11 Metadata [74,6 kB]
Des:11 http://security.ubuntu.com/ubuntu focal-security/main DEP-11 48x48 Icons [27,5 kB]
Des:12 http://security.ubuntu.com/ubuntu focal-security/main DEP-11 64x64 Icons [47,5 kB]
Des:13 http://security.ubuntu.com/ubuntu focal-security/main amd64 c-n-f Metadata [14,4 kB]
Des:14 http://security.ubuntu.com/ubuntu focal-security/restricted i386 Packages [42,5 kB]
Des:15 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 Packages [3.768 kB]
Des:16 http://es.archive.ubuntu.com/ubuntu focal-updates/main Translation-en [601 kB]
Des:17 http://es.archive.ubuntu.com/ubuntu focal-updates/main amd64 DEP-11 Metadata [276 kB]
Des:18 http://es.archive.ubuntu.com/ubuntu focal-updates/main DEP-11 48x48 Icons [64,2 kB]
Des:19 http://es.archive.ubuntu.com/ubuntu focal-updates/main DEP-11 64x64 Icons [103 kB]
Des:20 http://es.archive.ubuntu.com/ubuntu focal-updates/main amd64 c-n-f Metadata [18,0 kB]
Des:21 http://es.archive.ubuntu.com/ubuntu focal-updates/restricted amd64 Packages [3.922 kB]
Des:22 http://security.ubuntu.com/ubuntu focal-security/restricted Translation-en [527 kB]
Des:23 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 DEP-11 Metadata [212 B]
Des:24 http://security.ubuntu.com/ubuntu focal-security/restricted DEP-11 48x48 Icons [29 B]
Des:25 http://security.ubuntu.com/ubuntu focal-security/restricted DEP-11 64x64 Icons [29 B]
Des:26 http://security.ubuntu.com/ubuntu focal-security/restricted DEP-11 64x64@2 Icons [29 B]
Des:27 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 c-n-f Metadata [584 B]
Des:28 http://security.ubuntu.com/ubuntu focal-security/universe i386 Packages [698 kB]
Des:29 http://security.ubuntu.com/ubuntu focal-security/universe amd64 Packages [1.040 kB]
Des:30 http://es.archive.ubuntu.com/ubuntu focal-updates/restricted i386 Packages [44,1 kB]
Des:31 http://es.archive.ubuntu.com/ubuntu focal-updates/restricted Translation-en [548 kB]
Des:32 http://es.archive.ubuntu.com/ubuntu focal-updates/restricted amd64 DEP-11 Metadata [212 B]
Des:33 http://es.archive.ubuntu.com/ubuntu focal-updates/restricted DEP-11 48x48 Icons [29 B]
Des:34 http://es.archive.ubuntu.com/ubuntu focal-updates/restricted DEP-11 64x64 Icons [29 B]
Des:35 http://es.archive.ubuntu.com/ubuntu focal-updates/restricted DEP-11 64x64@2 Icons [29 B]
Des:36 http://es.archive.ubuntu.com/ubuntu focal-updates/restricted amd64 c-n-f Metadata [604 B]
Des:37 http://es.archive.ubuntu.com/ubuntu focal-updates/universe i386 Packages [823 kB]
Des:38 http://es.archive.ubuntu.com/ubuntu focal-updates/universe amd64 Packages [1.262 kB]
Des:39 http://es.archive.ubuntu.com/ubuntu focal-updates/universe Translation-en [303 kB]
Des:40 http://es.archive.ubuntu.com/ubuntu focal-updates/universe amd64 DEP-11 Metadata [446 kB]
Des:41 http://es.archive.ubuntu.com/ubuntu focal-updates/universe DEP-11 48x48 Icons [297 kB]
Des:42 http://es.archive.ubuntu.com/ubuntu focal-updates/universe DEP-11 64x64 Icons [520 kB]
Des:43 http://es.archive.ubuntu.com/ubuntu focal-updates/universe amd64 c-n-f Metadata [29,3 kB]
Des:44 http://es.archive.ubuntu.com/ubuntu focal-updates/multiverse i386 Packages [8.440 B]
Des:45 http://es.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 Packages [29,7 kB]
Des:46 http://es.archive.ubuntu.com/ubuntu focal-updates/multiverse Translation-en [8.316 B]
```

```
Des:47 http://es.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 DEP-11 Metadata [940 B]
Des:48 http://es.archive.ubuntu.com/ubuntu focal-updates/multiverse DEP-11 48x48 Icons [1.867 B]
Des:49 http://es.archive.ubuntu.com/ubuntu focal-updates/multiverse DEP-11 64x64 Icons [2.497 B]
Des:50 http://es.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 c-n-f Metadata [688 B]
Des:51 http://es.archive.ubuntu.com/ubuntu focal-backports/main i386 Packages [36,1 kB]
Des:52 http://es.archive.ubuntu.com/ubuntu focal-backports/main amd64 Packages [45,7 kB]
Des:53 http://es.archive.ubuntu.com/ubuntu focal-backports/main Translation-en [16,3 kB]
Des:54 http://es.archive.ubuntu.com/ubuntu focal-backports/main amd64 DEP-11 Metadata [7.980 B]
Des:55 http://es.archive.ubuntu.com/ubuntu focal-backports/main DEP-11 48x48 Icons [7.156 B]
Des:56 http://es.archive.ubuntu.com/ubuntu focal-backports/main DEP-11 64x64 Icons [10,7 kB]
Des:57 http://es.archive.ubuntu.com/ubuntu focal-backports/main amd64 c-n-f Metadata [1.420 B]
Des:58 http://es.archive.ubuntu.com/ubuntu focal-backports/restricted amd64 DEP-11 Metadata [216 B]
Des:59 http://es.archive.ubuntu.com/ubuntu focal-backports/restricted DEP-11 48x48 Icons [29 B]
Des:60 http://es.archive.ubuntu.com/ubuntu focal-backports/restricted DEP-11 64x64 Icons [29 B]
Des:61 http://es.archive.ubuntu.com/ubuntu focal-backports/restricted DEP-11 64x64@2 Icons [29 B]
Des:62 http://security.ubuntu.com/ubuntu focal-security/universe Translation-en [221 kB]
Des:63 http://es.archive.ubuntu.com/ubuntu focal-backports/universe i386 Packages [13,8 kB]
Des:64 http://es.archive.ubuntu.com/ubuntu focal-backports/universe amd64 Packages [25,0 kB]
Des:65 http://es.archive.ubuntu.com/ubuntu focal-backports/universe Translation-en [16,3 kB]
Des:66 http://es.archive.ubuntu.com/ubuntu focal-backports/universe amd64 DEP-11 Metadata [30,6 kB]
Des:67 http://es.archive.ubuntu.com/ubuntu focal-backports/universe DEP-11 48x48 Icons [13,3 kB]
Des:68 http://es.archive.ubuntu.com/ubuntu focal-backports/universe DEP-11 64x64 Icons [22,7 kB]
Des:69 http://es.archive.ubuntu.com/ubuntu focal-backports/universe amd64 c-n-f Metadata [880 B]
Des:70 http://es.archive.ubuntu.com/ubuntu focal-backports/multiverse amd64 DEP-11 Metadata [212 B]
Des:71 http://es.archive.ubuntu.com/ubuntu focal-backports/multiverse DEP-11 48x48 Icons [29 B]
Des:72 http://es.archive.ubuntu.com/ubuntu focal-backports/multiverse DEP-11 64x64 Icons [29 B]
Des:73 http://security.ubuntu.com/ubuntu focal-security/universe amd64 DEP-11 Metadata [160 kB]
Des:74 http://es.archive.ubuntu.com/ubuntu focal-backports/multiverse DEP-11 64x64@2 Icons [29 B]
Des:75 http://security.ubuntu.com/ubuntu focal-security/universe DEP-11 48x48 Icons [95,4 kB]
Des:76 http://security.ubuntu.com/ubuntu focal-security/universe DEP-11 64x64 Icons [166 kB]
Des:77 http://security.ubuntu.com/ubuntu focal-security/universe amd64 c-n-f Metadata [22,4 kB]
Des:78 http://security.ubuntu.com/ubuntu focal-security/multiverse i386 Packages [7.204 B]
Des:79 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64 Packages [26,6 kB]
Des:80 http://security.ubuntu.com/ubuntu focal-security/multiverse Translation-en [6.448 B]
Des:81 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64 DEP-11 Metadata [940 B]
Des:82 http://security.ubuntu.com/ubuntu focal-security/multiverse DEP-11 48x48 Icons [1.867 B]
Des:83 http://security.ubuntu.com/ubuntu focal-security/multiverse DEP-11 64x64 Icons [2.497 B]
Des:84 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64 c-n-f Metadata [604 B]
Descargados 26,9 MB en 5s (5.116 kB/s)
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias
Leyendo la información de estado... Hecho
Se pueden actualizar 641 paquetes. Ejecute «apt list --upgradable» para verlos.
bigdata@bigdata-IMF:~$
```

Ahora instalamos **curl** y seguidamente **xlsx2csv**:

```
bigdata@bigdata-IMF:~$ sudo apt install curl -y
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias
Leyendo la información de estado... Hecho
Los paquetes indicados a continuación se instalaron de forma automática y ya no son necesarios.
  linux-headers-5.8.0-43-generic linux-hwe-5.8-headers-5.8.0-43 linux-image-5.8.0-43-generic linux-modules-5.
8.0-43-generic linux-modules-extra-5.8.0-43-generic
Utilice «sudo apt autoremove» para eliminarlos.
Se instalarán los siguientes paquetes adicionales:
  libcurl4
Se instalarán los siguientes paquetes NUEVOS:
  curl
Se actualizarán los siguientes paquetes:
  libcurl4
1 actualizados, 1 nuevos se instalarán, 0 para eliminar y 640 no actualizados.
Se necesita descargar 396 kB de archivos.
Se utilizarán 425 kB de espacio de disco adicional después de esta operación.
Des:1 http://es.archive.ubuntu.com/ubuntu focal-updates/main amd64 libcurl4 amd64 7.68.0-1ubuntu2.25 [235 kB]
Des:2 http://es.archive.ubuntu.com/ubuntu focal-updates/main amd64 curl amd64 7.68.0-1ubuntu2.25 [162 kB]
Descargados 396 kB en 0s (891 kB/s)
(Leyendo la base de datos ... 231096 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../libcurl4_7.68.0-1ubuntu2.25_amd64.deb ...
Desempaquetando libcurl4:amd64 (7.68.0-1ubuntu2.25) sobre (7.68.0-1ubuntu2.4) ...
Seleccionando el paquete curl previamente no seleccionado.
Preparando para desempaquetar .../curl_7.68.0-1ubuntu2.25_amd64.deb ...
Desempaquetando curl (7.68.0-1ubuntu2.25) ...
Configurando libcurl4:amd64 (7.68.0-1ubuntu2.25) ...
Configurando curl (7.68.0-1ubuntu2.25) ...
Procesando disparadores para man-db (2.9.1-1) ...
Procesando disparadores para libc-bin (2.31-0ubuntu9.2) ...
bigdata@bigdata-IMF:~$
```

```
bigdata@bigdata-IMF:~$ sudo apt-get install xlsx2csv -y
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias
Leyendo la información de estado... Hecho
Los paquetes indicados a continuación se instalaron de forma automática y ya no son necesarios.
  linux-headers-5.8.0-43-generic linux-hwe-5.8-headers-5.8.0-43
  linux-image-5.8.0-43-generic linux-modules-5.8.0-43-generic
  linux-modules-extra-5.8.0-43-generic
Utilice «sudo apt autoremove» para eliminarlos.
Se instalarán los siguientes paquetes NUEVOS:
  xlsx2csv
0 actualizados, 1 nuevos se instalarán, 0 para eliminar y 640 no actualizados.
Se necesita descargar 178 kB de archivos.
Se utilizarán 359 kB de espacio de disco adicional después de esta operación.
Des:1 http://es.archive.ubuntu.com/ubuntu focal/universe amd64 xlsx2csv all 1:0.7.6-1 [178 kB]
Descargados 178 kB en 0s (820 kB/s)
Seleccionando el paquete xlsx2csv previamente no seleccionado.
(Leyendo la base de datos ... 231103 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../xlsx2csv_1%3a0.7.6-1_all.deb ...
Desempaquetando xlsx2csv (1:0.7.6-1) ...
Configurando xlsx2csv (1:0.7.6-1) ...
Procesando disparadores para man-db (2.9.1-1) ...
bigdata@bigdata-IMF:~$
```

Editamos los archivos **mapred-site.xml** y **yarn-site.xml** para indicarle a Hadoop dónde están sus propios jars y cómo debe ejecutar **MapReduce** sobre **YARN**. Esto lo hacemos a través del terminal utilizando el editor **nano**, o bien editando el archivo directamente.

En ***mapred-site.xml*** añadimos una nueva propiedad que indica dónde encontrar las clases de **MapReduce**.

```
  GNU nano 4.8                                                    /home/bigdata/hadoop/etc/hadoop/mapred-site.xml
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
  Licensed under the Apache License, Version 2.0 (the "License");
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
  <name>mapreduce.framework.name</name>
  <value>yarn</value>
</property>
<property>
  <name>mapreduce.application.classpath</name>
  <value>
    /home/bigdata/hadoop/share/hadoop/mapreduce/*,
    /home/bigdata/hadoop/share/hadoop/mapreduce/lib/*
  </value>
</property>
</configuration>
```

Y en ***yarn-site.xml*** introducimos en la propiedad ***yarn.application.classpath*** las rutas necesarias para que cualquier job encuentre las clases necesarias.

```
  GNU nano 4.8                                                    /home/bigdata/hadoop/etc/hadoop/yarn-site.xml
<?xml version="1.0"?>
<!--
  Licensed under the Apache License, Version 2.0 (the "License");
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<configuration>
 <property>
        <name>yarn.nodemanager.aux-services</name>
        <value>mapreduce_shuffle</value>
 </property>
<property>
        <name>yarn.application.classpath</name>
        <value>
          /home/bigdata/hadoop/etc/hadoop,
          /home/bigdata/hadoop/share/hadoop/common/*,
          /home/bigdata/hadoop/share/hadoop/common/lib/*,
          /home/bigdata/hadoop/share/hadoop/hdfs/*,
          /home/bigdata/hadoop/share/hadoop/hdfs/lib/*,
          /home/bigdata/hadoop/share/hadoop/mapreduce/*,
          /home/bigdata/hadoop/share/hadoop/mapreduce/lib/*,
          /home/bigdata/hadoop/share/hadoop/yarn/*,
          /home/bigdata/hadoop/share/hadoop/yarn/lib/*
        </value>
</property>
</configuration>
```

Por último, también editamos el archivo ***hive-site.xml***. Por un lado, para que no verifique estrictamente el esquema del metascore, dado que de base no tiene creadas tablas internas, y así evitamos errores. Y por otro lado, para que **Hive** cree automáticamente las tablas internas del metascore si estás no existen, evitando de nuevo posibles errores.

```
  GNU nano 4.8          /home/bigdata/hive/conf/hive-site.xml          Modificado
<?xml version="1.0"?>
<configuration>

  <property>
    <name>hive.metastore.schema.verification</name>
    <value>false</value>
  </property>

  <property>
    <name>datanucleus.schema.autoCreateTables</name>
    <value>true</value>
  </property>

</configuration>

^G Ver ayuda   ^O Guardar     ^W Buscar      ^K Cortar Texto^J Justificar
^X Salir       ^R Leer fich.  ^\ Reemplazar  ^U Pegar       ^T Ortografía
```

Una vez configurado todo, iniciamos **HDFS** y **YARN**, y verificamos que los procesos necesarios se están ejecutando.

```
bigdata@bigdata-IMF:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bigdata-IMF]
bigdata@bigdata-IMF:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
bigdata@bigdata-IMF:~$ jps
47298 DataNode
49059 Jps
47525 SecondaryNameNode
48778 NodeManager
47146 NameNode
48299 ResourceManager
bigdata@bigdata-IMF:~$
```

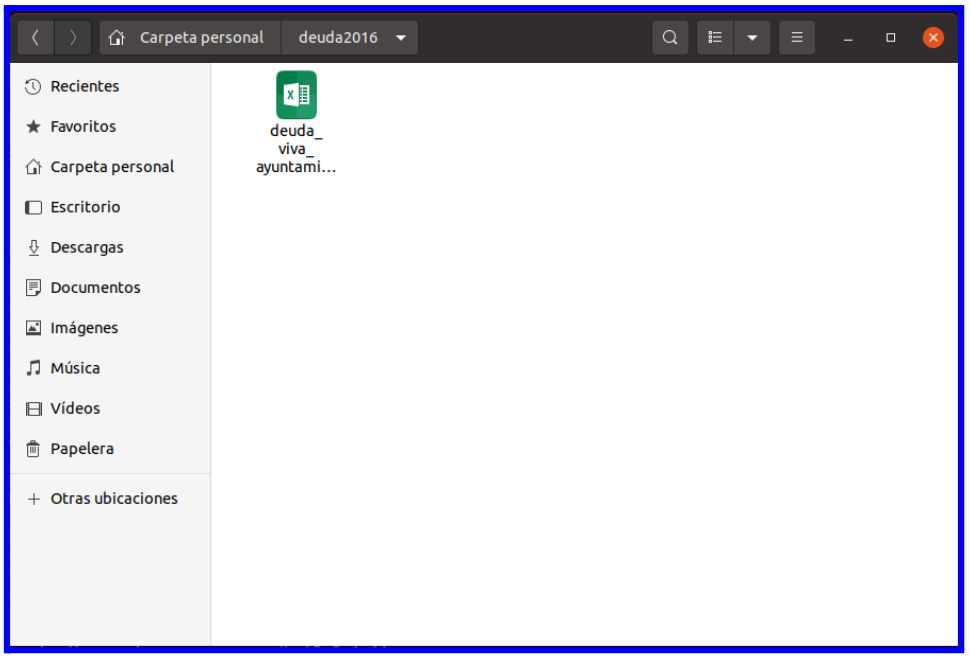
1. Descarga y conversión del fichero:

    a. Creamos la carpeta ***deuda2016*** y movemos el directorio a esta.

```
bigdata@bigdata-IMF:~$ mkdir -p ~/deuda2016 && cd ~/deuda2016
bigdata@bigdata-IMF:~/deuda2016$
```

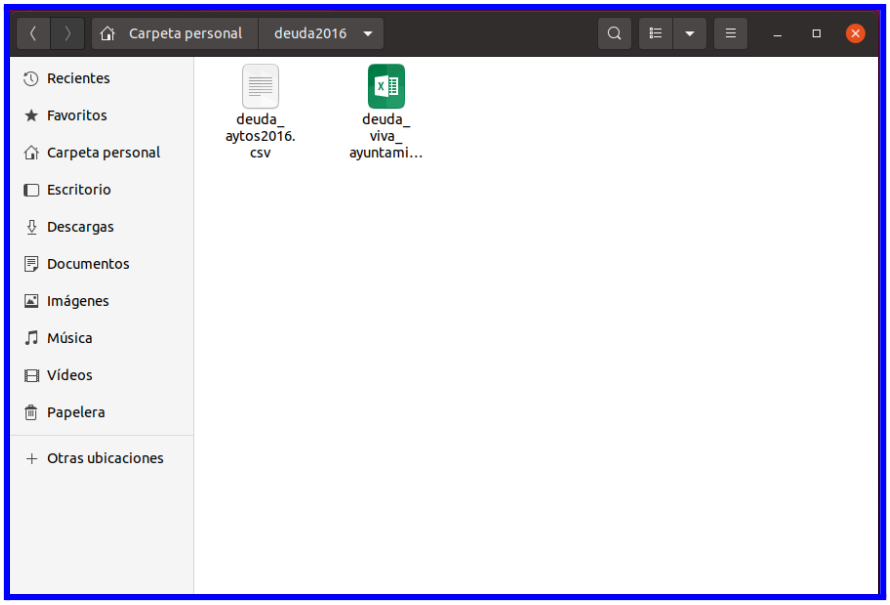
    b. Utilizando **curl** y con el enlace facilitado, descargamos el fichero.

```
bigdata@bigdata-IMF:~/deuda2016$ curl -L -o deuda_viva_ayuntamientos_2016.xlsx \
>   "https://www.hacienda.gob.es/CDI/sist%20financiacion%20y%20deuda/informacioneells/2016/deuda_viva_ayuntamientos_2016__ovel_20170630.xlsx"
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 1027k  100 1027k    0     0  3003k      0 --:--:-- --:--:-- --:--:-- 3003k
bigdata@bigdata-IMF:~/deuda2016$
```



c. Utilizando **xlsx2csv**, convertimos el fichero a .csv.

```
bigdata@bigdata-IMF:~/deuda2016$ xlsx2csv deuda_viva_ayuntamientos_2016.xlsx > deuda_aytos2016.csv
bigdata@bigdata-IMF:~/deuda2016$
```



d. Por último, limpiamos las líneas vacías o espacios, y comprobamos las primeras diez líneas.

```
bigdata@bigdata-IMF:~/deuda2016$ sed -i '/^[[:space:]]*$/d' deuda_aytos2016.csv
bigdata@bigdata-IMF:~/deuda2016$
```

```
bigdata@bigdata-IMF:~/deuda2016$ head -n 10 deuda_aytos2016.csv
Ejercicio,Código     CC.AA.,CC.AA.,Código Provincia,Provincia,Código Municipio,Municipio,"Deuda viva    31/12/2016
    (miles de euros)",,
Ejercicio,cdcdad,nomcdad010,cdprov,nomprov010,cdcorp,nomcorp,ImpDeudViva,,
2016,01,ANDALUCIA,04,ALMERIA,001,Abla                                                                  ,359.356,,
2016,01,ANDALUCIA,04,ALMERIA,002,Abrucena                                                              ,25.104,,
2016,01,ANDALUCIA,04,ALMERIA,003,Adra                                                                  ,17160.83552,,
2016,01,ANDALUCIA,04,ALMERIA,004,Albánchez                                                             ,76.607,,
2016,01,ANDALUCIA,04,ALMERIA,005,Alboloduy                                                             ,0,,
2016,01,ANDALUCIA,04,ALMERIA,006,Albox                                                                 ,20597.51244,,
2016,01,ANDALUCIA,04,ALMERIA,007,Alcolea                                                               ,278.20894,,
bigdata@bigdata-IMF:~/deuda2016$
```

2. Crear carpeta en **HDFS** y subir el fichero:

a. Creamos la carpeta y subimos el fichero.

```
bigdata@bigdata-IMF:~/deuda2016$ hdfs dfs -mkdir -p /user/bigdata/deuda2016
bigdata@bigdata-IMF:~/deuda2016$ hdfs dfs -put -f ~/deuda2016/deuda_aytos2016.csv /user/bigdata/deuda2016/
bigdata@bigdata-IMF:~/deuda2016$
```

b. Comprobamos que el fichero se encuentra en la carpeta **HDFS**.

```
bigdata@bigdata-IMF:~/deuda2016$ hdfs dfs -ls /user/bigdata/deuda2016
Found 1 items
-rw-r--r--   1 bigdata supergroup     925336 2025-11-26 14:17 /user/bigdata/deuda2016/deuda_aytos2016.csv
bigdata@bigdata-IMF:~/deuda2016$
```

3. Creación de la tabla en **Hive**:

   a. Antes de iniciar **Hive**, movemos el directorio a la carpeta **Hive** e inicializamos el metascore.

```
bigdata@bigdata-IMF:~/hive$ bin/schematool -dbType derby -initSchema
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/bigdata/hive/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/bigdata/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.30.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Metastore connection URL:        jdbc:derby:;databaseName=metastore_db;create=true
Metastore Connection Driver :    org.apache.derby.jdbc.EmbeddedDriver
Metastore connection User:       APP
Starting metastore schema initialization to 3.1.0
Initialization script hive-schema-3.1.0.derby.sql
```

```
Initialization script completed
schemaTool completed
bigdata@bigdata-IMF:~/hive$
```

   b. Ahora si, iniciamos **Hive**.

```
bigdata@bigdata-IMF:~/hive$ hive
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/bigdata/hive/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/bigdata/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.30.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Hive Session ID = e396e60e-7a05-4b00-bbf2-a16b10db6181

Logging initialized using configuration in jar:file:/home/bigdata/hive/lib/hive-common-3.1.2.jar!/hive-log4j2.properties Async: true
2025-11-26 18:20:41,292 INFO  [e396e60e-7a05-4b00-bbf2-a16b10db6181 main] DataNucleus.Persistence: Propiedad datanucleus.cache.level2 desconocida - vamos a ignorarla
2025-11-26 18:20:45,545 WARN  [e396e60e-7a05-4b00-bbf2-a16b10db6181 main] DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
2025-11-26 18:20:45,552 WARN  [e396e60e-7a05-4b00-bbf2-a16b10db6181 main] DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
2025-11-26 18:20:45,553 WARN  [e396e60e-7a05-4b00-bbf2-a16b10db6181 main] DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
2025-11-26 18:20:45,553 WARN  [e396e60e-7a05-4b00-bbf2-a16b10db6181 main] DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
2025-11-26 18:20:45,553 WARN  [e396e60e-7a05-4b00-bbf2-a16b10db6181 main] DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
2025-11-26 18:20:45,554 WARN  [e396e60e-7a05-4b00-bbf2-a16b10db6181 main] DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
2025-11-26 18:20:47,483 WARN  [e396e60e-7a05-4b00-bbf2-a16b10db6181 main] DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
2025-11-26 18:20:47,485 WARN  [e396e60e-7a05-4b00-bbf2-a16b10db6181 main] DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
2025-11-26 18:20:47,486 WARN  [e396e60e-7a05-4b00-bbf2-a16b10db6181 main] DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
2025-11-26 18:20:47,486 WARN  [e396e60e-7a05-4b00-bbf2-a16b10db6181 main] DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
2025-11-26 18:20:47,492 WARN  [e396e60e-7a05-4b00-bbf2-a16b10db6181 main] DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
2025-11-26 18:20:47,492 WARN  [e396e60e-7a05-4b00-bbf2-a16b10db6181 main] DataNucleus.MetaData: Metadata has jdbc-type of null yet this is not valid. Ignored
Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X releases.
Hive Session ID = 2613e0ac-8d4c-4863-8b68-31e8a803752f
hive>
```

   c. A continuación, creamos las tablas. Se ha optado por crear una tabla “raw” y después una tabla limpia. Esto se debe a que al usar **OpenCSVSerde** la variable o columna que contiene las deudas era automáticamente transformada en **string** y no se establecía como **double**. Sin embargo, al optar por no utilizar **OpenCSVSerde**, había problemas en la separación de columnas con filas como RIOJA, LA.

   d. Por tanto, primero se crea una **tabla raw** con las distintas variables y haciendo uso de **OpenCSVSerde** y su adecuado separador.

```
hive> CREATE EXTERNAL TABLE deuda_aytos_2016_raw (
    >   Anio            INT,
    >   Cod_Comunidad   STRING,
    >   Comunidad       STRING,
    >   Cod_Provincia   STRING,
    >   Provincia       STRING,
    >   Cod_corp        STRING,
    >   Corporacion     STRING,
    >   Deuda_str       STRING,
    >   extra1          STRING,
    >   extra2          STRING
    > )
    > ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.OpenCSVSerde'
    > WITH SERDEPROPERTIES (
    >   "separatorChar" = ",",
    >   "quoteChar"     = "\""
    > )
    > STORED AS TEXTFILE
    > LOCATION '/user/bigdata/deuda2016/'
    > TBLPROPERTIES ("skip.header.line.count"="2");
OK
Time taken: 0.104 seconds
hive>
```

e. Después se crea una **tabla limpia** con las mismas variables pero transformando la columna **Deuda_str** en una nueva con tipo **double**.

```
hive> CREATE TABLE deuda_aytos_2016 AS
    > SELECT
    >   Anio,
    >   Cod_Comunidad,
    >   Comunidad,
    >   Cod_Provincia,
    >   Provincia,
    >   Cod_corp,
    >   Corporacion,
    >   CAST(Deuda_str AS DOUBLE) AS Deuda
    > FROM deuda_aytos_2016_raw;
```

```
hive> CREATE TABLE deuda_aytos_2016 AS
    > SELECT
    >   Anio,
    >   Cod_Comunidad,
    >   Comunidad,
    >   Cod_Provincia,
    >   Provincia,
    >   Cod_corp,
    >   Corporacion,
    >   CAST(Deuda_str AS DOUBLE) AS Deuda
    > FROM deuda_aytos_2016_raw;
Query ID = bigdata_20251202151249_8cf756f2-cea6-4107-8be6-88f8bdcef596
Total jobs = 3
Launching Job 1 out of 3
Number of reduce tasks is set to 0 since there's no reduce operator
Starting Job = job_1764197870109_0044, Tracking URL = http://bigdata-IMF:8088/proxy/application_1764197870109_0044/
Kill Command = /home/bigdata/hadoop/bin/mapred job  -kill job_1764197870109_0044
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 0
2025-12-02 15:12:57,185 Stage-1 map = 0%,  reduce = 0%
2025-12-02 15:13:03,367 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 1.49 sec
MapReduce Total cumulative CPU time: 1 seconds 490 msec
Ended Job = job_1764197870109_0044
Stage-4 is selected by condition resolver.
Stage-3 is filtered out by condition resolver.
Stage-5 is filtered out by condition resolver.
Moving data to directory hdfs://localhost:9000/user/hive/warehouse/.hive-staging_hive_2025-12-02_15-12-49_529_6970142413360383354-1/-ext-10002
Moving data to directory hdfs://localhost:9000/user/hive/warehouse/deuda_aytos_2016
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1   Cumulative CPU: 1.49 sec   HDFS Read: 931650 HDFS Write: 915645 SUCCESS
Total MapReduce CPU Time Spent: 1 seconds 490 msec
OK
Time taken: 15.107 seconds
hive>
```

f. Por último comprobamos que la tabla se haya creado correctamente.

```
hive> SELECT * FROM deuda_aytos_2016 LIMIT 10;
OK
Ejercicio       cdcdad  nomcdad010      cdprov  nomprov010      cdcorp  nomcorp NULL
2016    01      ANDALUCIA       04      ALMERIA 001     Abla                                                    359.356
2016    01      ANDALUCIA       04      ALMERIA 002     Abrucena                                                25.104
2016    01      ANDALUCIA       04      ALMERIA 003     Adra                                                    17160.83552
2016    01      ANDALUCIA       04      ALMERIA 004     Albánchez                                               76.607
2016    01      ANDALUCIA       04      ALMERIA 005     Alboloduy                                               0.0
2016    01      ANDALUCIA       04      ALMERIA 006     Albox                                                   20597.51244
2016    01      ANDALUCIA       04      ALMERIA 007     Alcolea                                                 278.20894
2016    01      ANDALUCIA       04      ALMERIA 008     Alcóntar                                                179.69309
2016    01      ANDALUCIA       04      ALMERIA 009     Alcudia de Monteagud                                    27.39363
Time taken: 0.167 seconds, Fetched: 10 row(s)
hive>
```

4. Consultas solicitadas:

a. Corporación que tiene la deuda más grande.

```
hive> WITH orden AS (
    >   SELECT Corporacion, Deuda,
    >          ROW_NUMBER() OVER (ORDER BY Deuda DESC) AS rn
    >   FROM deuda_aytos_2016
    >   WHERE UPPER(TRIM(Corporacion)) <> 'TOTAL'
    > )
    > SELECT *
    > FROM orden
    > WHERE rn = 3;
```

```
hive> WITH orden AS (
    >   SELECT Corporacion, Deuda,
    >          ROW_NUMBER() OVER (ORDER BY Deuda DESC) AS rn
    >   FROM deuda_aytos_2016
    >   WHERE UPPER(TRIM(Corporacion)) <> 'TOTAL'
    > )
    > SELECT *
    > FROM orden
    > WHERE rn = 3;
Query ID = bigdata_20251202151523_fc9ba587-dcb2-4d78-a040-5053b81b45b2
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1764197870109_0045, Tracking URL = http://bigdata-IMF:8088/proxy/application_1764197870109_0045/
Kill Command = /home/bigdata/hadoop/bin/mapred job  -kill job_1764197870109_0045
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2025-12-02 16:13:55,335 Stage-1 map = 0%,  reduce = 0%
2025-12-02 16:14:01,500 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 1.76 sec
2025-12-02 16:14:08,718 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 3.69 sec
MapReduce Total cumulative CPU time: 3 seconds 690 msec
Ended Job = job_1764197870109_0045
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 3.69 sec   HDFS Read: 931904 HDFS Write: 181 SUCCESS
Total MapReduce CPU Time Spent: 3 seconds 690 msec
OK
Barcelona                                                          840000.0        3
Time taken: 3527.165 seconds, Fetched: 1 row(s)
hive>
```

Barcelona

b. Corporación que tiene la deuda más grande de Andalucía.

```
hive> SELECT Corporacion, Deuda
    > FROM deuda_aytos_2016
    > WHERE UPPER(Comunidad) = 'ANDALUCIA'
    > ORDER BY Deuda DESC
    > LIMIT 1;
```

```
hive> SELECT Corporacion, Deuda
    > FROM deuda_aytos_2016
    > WHERE UPPER(Comunidad) = 'ANDALUCIA'
    > ORDER BY Deuda DESC
    > LIMIT 1;
Query ID = bigdata_20251202161540_deff24ed-aaad-421b-bf7a-abe2a140517a
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1764197870109_0046, Tracking URL = http://bigdata-IMF:8088/proxy/application_1764197870109_0046/
Kill Command = /home/bigdata/hadoop/bin/mapred job  -kill job_1764197870109_0046
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2025-12-02 16:15:47,726 Stage-1 map = 0%,  reduce = 0%
2025-12-02 16:15:53,894 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 1.6 sec
2025-12-02 16:16:00,075 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 2.96 sec
MapReduce Total cumulative CPU time: 2 seconds 960 msec
Ended Job = job_1764197870109_0046
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 2.96 sec   HDFS Read: 928931 HDFS Write: 181 SUCCESS
Total MapReduce CPU Time Spent: 2 seconds 960 msec
OK
Jerez de la Frontera                                               636885.676
Time taken: 22.071 seconds, Fetched: 1 row(s)
hive>
```

```
Jerez de la Frontera
```

c. Suma de deuda total de cada comunidad autónoma, ordenada por comunidad.

```
hive> SELECT Comunidad, SUM(Deuda) AS total_deuda
    > FROM deuda_aytos_2016
    > GROUP BY Comunidad
    > ORDER BY Comunidad;
```

```
hive> SELECT Comunidad, SUM(Deuda) AS total_deuda
    > FROM deuda_aytos_2016
    > GROUP BY Comunidad
    > ORDER BY Comunidad;
Query ID = bigdata_20251202161826_5af4c03b-800c-4db4-806b-036dfe020a30
Total jobs = 2
Launching Job 1 out of 2
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1764197870109_0047, Tracking URL = http://bigdata-IMF:8088/proxy/application_1764197870109_0047/
Kill Command = /home/bigdata/hadoop/bin/mapred job  -kill job_1764197870109_0047
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2025-12-02 16:18:34,491 Stage-1 map = 0%,  reduce = 0%
2025-12-02 16:18:39,699 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 1.24 sec
2025-12-02 16:18:45,867 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 2.42 sec
MapReduce Total cumulative CPU time: 2 seconds 420 msec
Ended Job = job_1764197870109_0047
Launching Job 2 out of 2
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1764197870109_0048, Tracking URL = http://bigdata-IMF:8088/proxy/application_1764197870109_0048/
Kill Command = /home/bigdata/hadoop/bin/mapred job  -kill job_1764197870109_0048
Hadoop job information for Stage-2: number of mappers: 1; number of reducers: 1
2025-12-02 16:18:58,597 Stage-2 map = 0%,  reduce = 0%
2025-12-02 16:19:04,764 Stage-2 map = 100%,  reduce = 0%, Cumulative CPU 1.07 sec
2025-12-02 16:19:09,941 Stage-2 map = 100%,  reduce = 100%, Cumulative CPU 2.23 sec
MapReduce Total cumulative CPU time: 2 seconds 230 msec
Ended Job = job_1764197870109_0048
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 2.42 sec   HDFS Read: 929308 HDFS Write: 810 SUCCESS
Stage-Stage-2: Map: 1  Reduce: 1   Cumulative CPU: 2.23 sec   HDFS Read: 8379 HDFS Write: 887 SUCCESS
Total MapReduce CPU Time Spent: 4 seconds 650 msec
```

```
OK
        2.6142561120961E7
ANDALUCIA       6258833.119060001
ARAGON  1176398.883458
ASTURIAS        325280.62334999995
C.VALENCIANA    2485272.1670900006
CANARIAS        506214.7844399999
CANTABRIA       127052.61321999998
CASTILLA-LEON   1055232.6923600005
CASTILLA-MANCHA 801413.8976499999
CATALUÑA        4181548.7130200006
CEUTA   202279.32681
EXTREMADURA     251298.0152899999
GALICIA 420727.7954200004
ILLES BALEARS   561621.9769299999
MADRID  6111572.4654300045
MELILLA 94186.262
MURCIA  923845.1972620003
NAVARRA 211208.26217
PAIS VASCO      463612.00399999984
RIOJA   66670.32199999997
nomcdad010      NULL
Time taken: 45.568 seconds, Fetched: 21 row(s)
hive>
```

d. Suma de deuda de corporaciones que empiezan por vocal.

```
hive> SELECT SUM(Deuda)
    > FROM deuda_aytos_2016
    > WHERE UPPER(SUBSTR(Corporacion, 1, 1)) IN ('A','E','I','O','U');
```

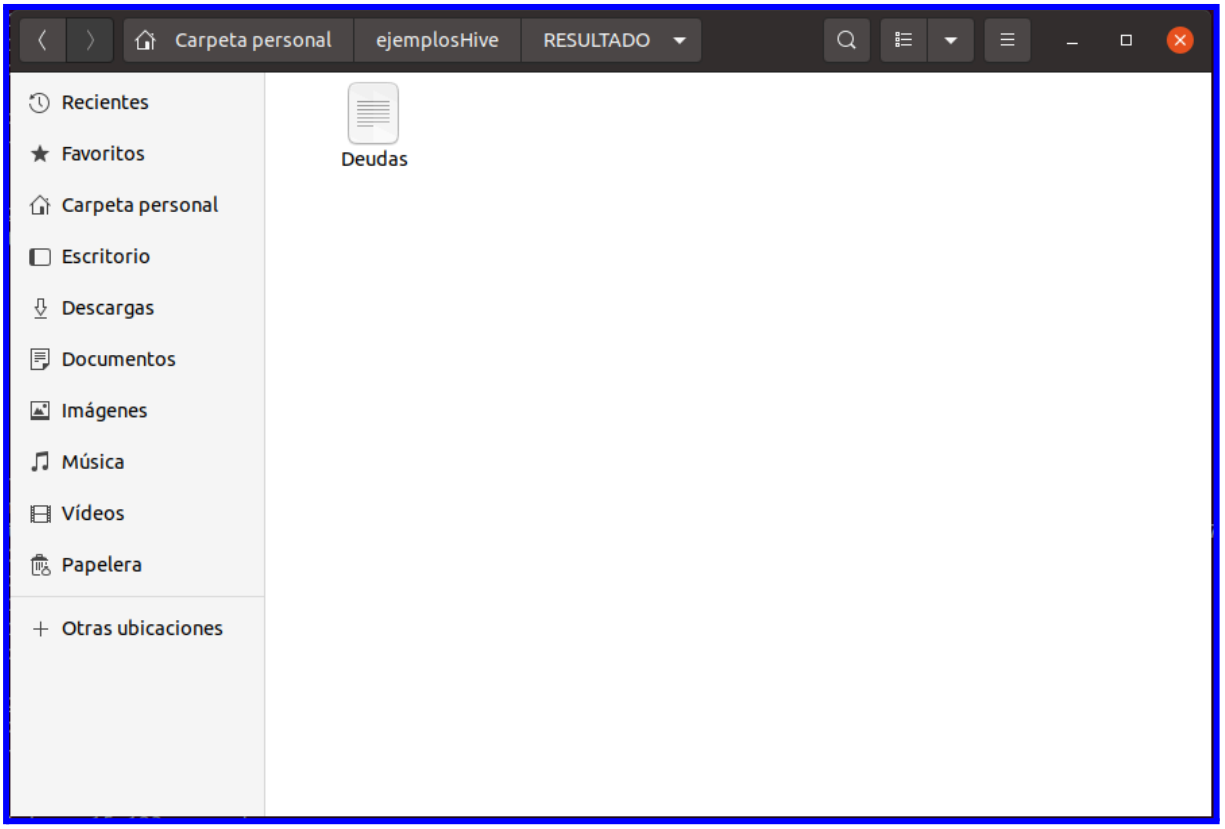
```
hive> SELECT SUM(Deuda)
    > FROM deuda_aytos_2016
    > WHERE UPPER(SUBSTR(Corporacion, 1, 1)) IN ('A','E','I','O','U');
Query ID = bigdata_20251202162127_6c3464cd-f48b-4a27-a53e-c7827dbcd95d
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1764197870109_0049, Tracking URL = http://bigdata-IMF:8088/proxy/application_1764197870109_0049/
Kill Command = /home/bigdata/hadoop/bin/mapred job  -kill job_1764197870109_0049
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2025-12-02 16:21:34,684 Stage-1 map = 0%,  reduce = 0%
2025-12-02 16:21:40,967 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 1.61 sec
2025-12-02 16:21:48,148 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 2.96 sec
MapReduce Total cumulative CPU time: 2 seconds 960 msec
Ended Job = job_1764197870109_0049
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 2.96 sec   HDFS Read: 932142 HDFS Write: 117 SUCCESS
Total MapReduce CPU Time Spent: 2 seconds 960 msec
OK
3158922.465520005
Time taken: 21.287 seconds, Fetched: 1 row(s)
hive>
```

3158922.465520005

e. Exportar las corporaciones cuya deuda sea mayor a 10 000 y pertenezcan a Cataluña, a un fichero llamado ***Deudas***.

```
hive> INSERT OVERWRITE LOCAL DIRECTORY '/home/bigdata/ejemplosHive/RESULTADO/'
    > ROW FORMAT DELIMITED
    > FIELDS TERMINATED BY ','
    > SELECT *
    > FROM deuda_aytos_2016
    > WHERE Comunidad = 'CATALUÑA'
    >   AND Deuda > 10000;
```

```
hive> INSERT OVERWRITE LOCAL DIRECTORY '/home/bigdata/ejemplosHive/RESULTADO/'
    > ROW FORMAT DELIMITED
    > FIELDS TERMINATED BY ','
    > SELECT *
    > FROM deuda_aytos_2016
    > WHERE Comunidad = 'CATALUÑA'
    >   AND Deuda > 10000;
Query ID = bigdata_20251202162252_eebc616e-f93a-4d5e-9107-b0d9fc27b448
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks is set to 0 since there's no reduce operator
Starting Job = job_1764197870109_0050, Tracking URL = http://bigdata-IMF:8088/proxy/application_1764197870109_0050/
Kill Command = /home/bigdata/hadoop/bin/mapred job  -kill job_1764197870109_0050
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 0
2025-12-02 16:22:58,952 Stage-1 map = 0%,  reduce = 0%
2025-12-02 16:23:06,142 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 1.77 sec
MapReduce Total cumulative CPU time: 1 seconds 770 msec
Ended Job = job_1764197870109_0050
Moving data to local directory /home/bigdata/ejemplosHive/RESULTADO
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1   Cumulative CPU: 1.77 sec   HDFS Read: 922633 HDFS Write: 7968 SUCCESS
Total MapReduce CPU Time Spent: 1 seconds 770 msec
OK
Time taken: 15.123 seconds
hive>
```



```
1 2016,09,CATALUÑA,08,BARCELONA,015,Badalona ,74860.20027
2 2016,09,CATALUÑA,08,BARCELONA,019,Barcelona ,840000.0
3 2016,09,CATALUÑA,08,BARCELONA,022,Berga ,14237.0589
4 2016,09,CATALUÑA,08,BARCELONA,051,Castellar del Vallès ,12324.96931
5 2016,09,CATALUÑA,08,BARCELONA,056,Castelldefels ,33646.972
6 2016,09,CATALUÑA,08,BARCELONA,073,Cornellà de Llobregat ,58238.96178
7 2016,09,CATALUÑA,08,BARCELONA,077,Esplugues de Llobregat ,15523.86602
8 2016,09,CATALUÑA,08,BARCELONA,086,Franqueses del Vallès (Les) ,10570.24559
9 2016,09,CATALUÑA,08,BARCELONA,089,Gavà ,33980.22376
10 2016,09,CATALUÑA,08,BARCELONA,096,Granollers ,26905.61337
11 2016,09,CATALUÑA,08,BARCELONA,101,Hospitalet de Llobregat (L') ,96833.41589
12 2016,09,CATALUÑA,08,BARCELONA,102,Igualada ,38842.5102
13 2016,09,CATALUÑA,08,BARCELONA,112,Manlleu ,11260.16779
14 2016,09,CATALUÑA,08,BARCELONA,113,Manresa ,72428.18509
15 2016,09,CATALUÑA,08,BARCELONA,114,Martorell ,13178.428
16 2016,09,CATALUÑA,08,BARCELONA,121,Mataró ,122683.46657
17 2016,09,CATALUÑA,08,BARCELONA,123,Molins de Rei ,16219.74991
18 2016,09,CATALUÑA,08,BARCELONA,124,Mollet del Vallès ,40581.79391
19 2016,09,CATALUÑA,08,BARCELONA,125,Montcada i Reixac ,16398.63819
20 2016,09,CATALUÑA,08,BARCELONA,138,Moià ,10822.92291
21 2016,09,CATALUÑA,08,BARCELONA,163,Pineda de Mar ,10973.37711
22 2016,09,CATALUÑA,08,BARCELONA,184,Rubí ,26942.19839
23 2016,09,CATALUÑA,08,BARCELONA,187,Sabadell ,97066.24229
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